Missing and Unbalanced Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected sampling-and-coverage audit.

Name the target, distinguish file counts, inspect crossed conditions, propose the sound plan and audit missing outcomes.

Keep one habit: Keep target, frame, selected cases and assessed results distinguishable.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Good Data, Bad Data

Similarity

Boundaries and 'Not Sure'

W1 · Name target, sample and missing conditions

> Use the supplied Shapes24 reference.

Your evidence record

- Shapes24 reference: six distinct cards in each of bright circles, dim circles, bright squares and dim squares. The original sample is all six bright circles; five recognition outputs were correct and one wrong. The original selection frame contains only bright-circle IDs. The other eighteen cards have not been assessed.

- Use the 24-card investigation. State the target collection, the original six-card **sample** and the three absent conditions.

  If the selection list contains only bright-circle IDs, explain why random selection from that list cannot reach every target condition.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Separate files from distinct cases

> Keep the underlying card IDs with the file count.

Your evidence record

- Each of the six original **sample**d cards is photographed twice. State the number of image files and distinct cards. Has this covered a new shape or lighting condition? Explain why the counts need different labels.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Check crossed combinations

> Name absent cells even when both individual labels appear.

Your evidence record

- A different test includes 3 bright circles and 3 dim squares. A designer says it covers both shapes and both lighting conditions, so no combinations are missing.

  Identify the missing combinations and propose six additional distinct cards to investigate them.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit a new sound-recognition claim

> Keep one first attempt before hints or feedback.

Read and explain independently

A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit a new sound-recognition claim

MAIN LESSON · SECTION 16

W5 · Audit the selected sound plan after the first attempt

> Retain the W4 plan and report its unavailable recognition file separately.

Your evidence record

- An 8-clip plan selects 2 from each sound condition. One selected noisy-clap file cannot be opened. The task measures recognition correctness only. State selected and assessed counts and the assessed count for each condition.

  Explain why a report of all 8 checked is wrong, and how to handle the missing file transparently.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

List normal, low-light, partly hidden, worn, and unusual items; mark which conditions are missing from the sorter data.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Separate files from distinct cases for Sampling and **coverage**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Sampling and **coverage**. Proposed explanation: “More records always remove bias”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A **training** set has 18 daylight photos and two dark photos of one object. Adding eight extra copies of each dark photo makes 18 dark-image files, matching the 18 daylight files. Does it balance the variety? Design a better collection step.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Training includes nine daylight leaf photos and one low-light photo.

What intended-use gap is present?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A photo sorter trains on ten bright photos and no dim photos. What group of examples is missing? Design a test that reveals the gap.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 22 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A **training** set has 18 daylight photos and two dark photos of one object. Adding eight extra copies of each dark photo makes 18 dark-image files, matching the 18 daylight files. Does it balance the variety? Design a better collection step.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional badge reader will be used in both bright and dim rooms.

Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

AI ACADEMY / AI-1/22

Target groups and development coverage

	Target cards	Sample cards
Bright plain	8	6
Bright patterned	8	2
Dim plain	8	0
Dim patterned	8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/22

Missing and Unbalanced Examples

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Target conditions

Four lighting-by-pattern groups, eight target cards each.



2. Current sample

Eight bright cards only; dim groups have zero coverage.



3. Coverage gap

Current results cannot describe dim-room reading.



4. Targeted follow-up

Select and record new cases from both dim groups before claiming coverage.

CHANGE ONE THING
Add the two remaining bright plain cards and no dim cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Would testing only dim plain cards close every current gap? Point to the target groups.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional badge reader will be used in both bright and dim rooms.
- Which labelled step matches this detail? Select and record new cases from both dim groups before claiming **coverage**.
- 1. Target conditions
- 2. Current **sample**
- 3. Targeted follow-up
- 4. **Coverage gap**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Targeted follow-up Select and record new cases from both dim groups before claiming **coverage**.

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Changed-case question

- Changed-case practice: Add the two remaining bright plain cards and no dim cards. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The **sample** becomes larger but still has no dim **coverage**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The **sample** becomes larger but still has no dim **coverage**. The missing lighting condition remains untested.

Repair and save your work

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Missing and Unbalanced Examples: the case

AI ACADEMY / AI-1/22 · 01 · LOOK AT THE FACTS

Missing and Unbalanced Examples: the case

The situation

A fictional badge reader will be used in both bright and dim rooms.

Supplied evidence

Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards. A development sample contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

By the end, I can identify missing conditions and design a targeted coverage check.

Follow the labelled picture with the matching explanation. Open the vector diagram

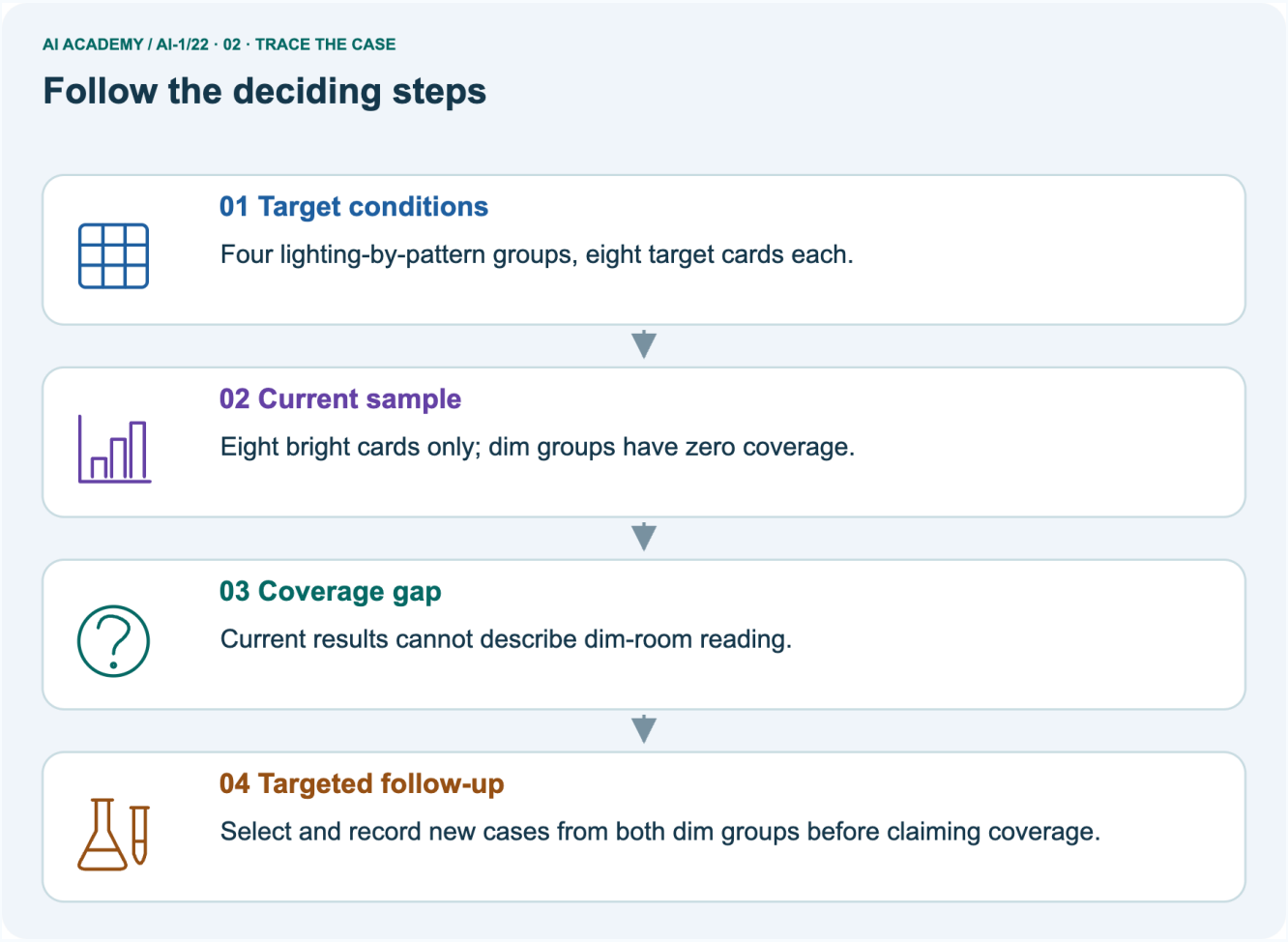
A fictional badge reader will be used in both bright and dim rooms. Supplied facts: Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target conditions: Four lighting-by-pattern groups, eight target cards each. → Current **sample**: Eight bright cards only; dim groups have zero **coverage**. → **Coverage gap**: Current results cannot describe dim-room reading. → Targeted follow-up: Select and record new cases from both dim groups before claiming **coverage**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Target groups and development coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

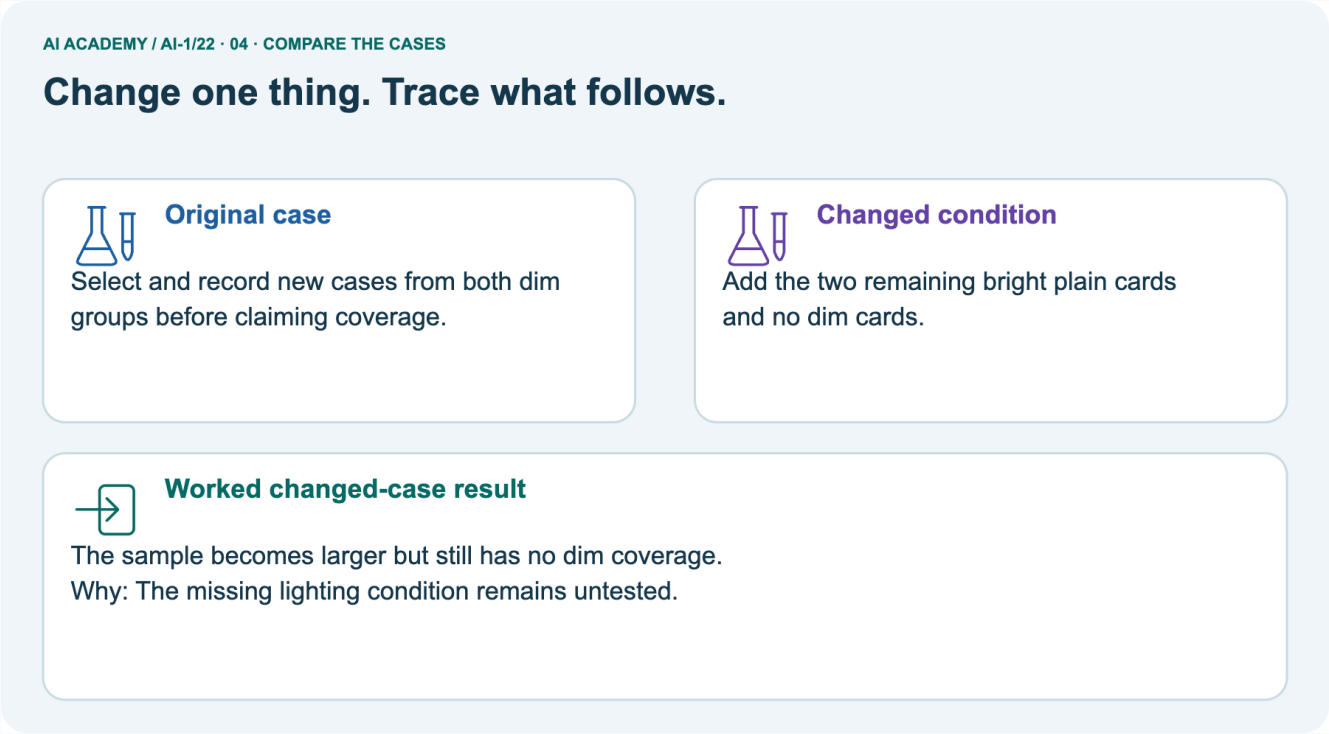
Target groups and development **coverage**. Bright plain: Target cards = 8, **Sample** cards = 6. Bright patterned: Target cards = 8, **Sample** cards = 2. Dim plain: Target cards = 8, **Sample** cards = 0.

Dim patterned: Target cards = 8, **Sample** cards = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

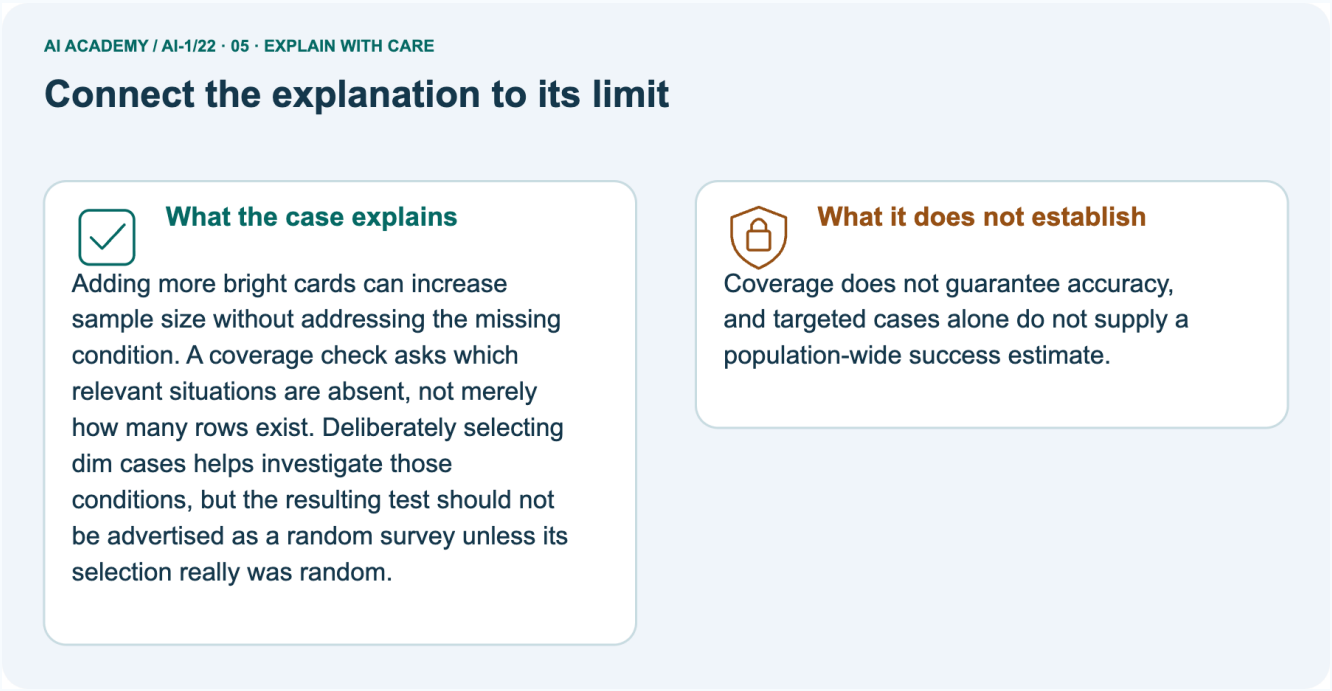
Original case: Select and record new cases from both dim groups before claiming **coverage**.
Changed condition: Add the two remaining bright plain cards and no dim cards.

Expected result: The **sample** becomes larger but still has no dim **coverage**. Reason: The missing lighting condition remains untested.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

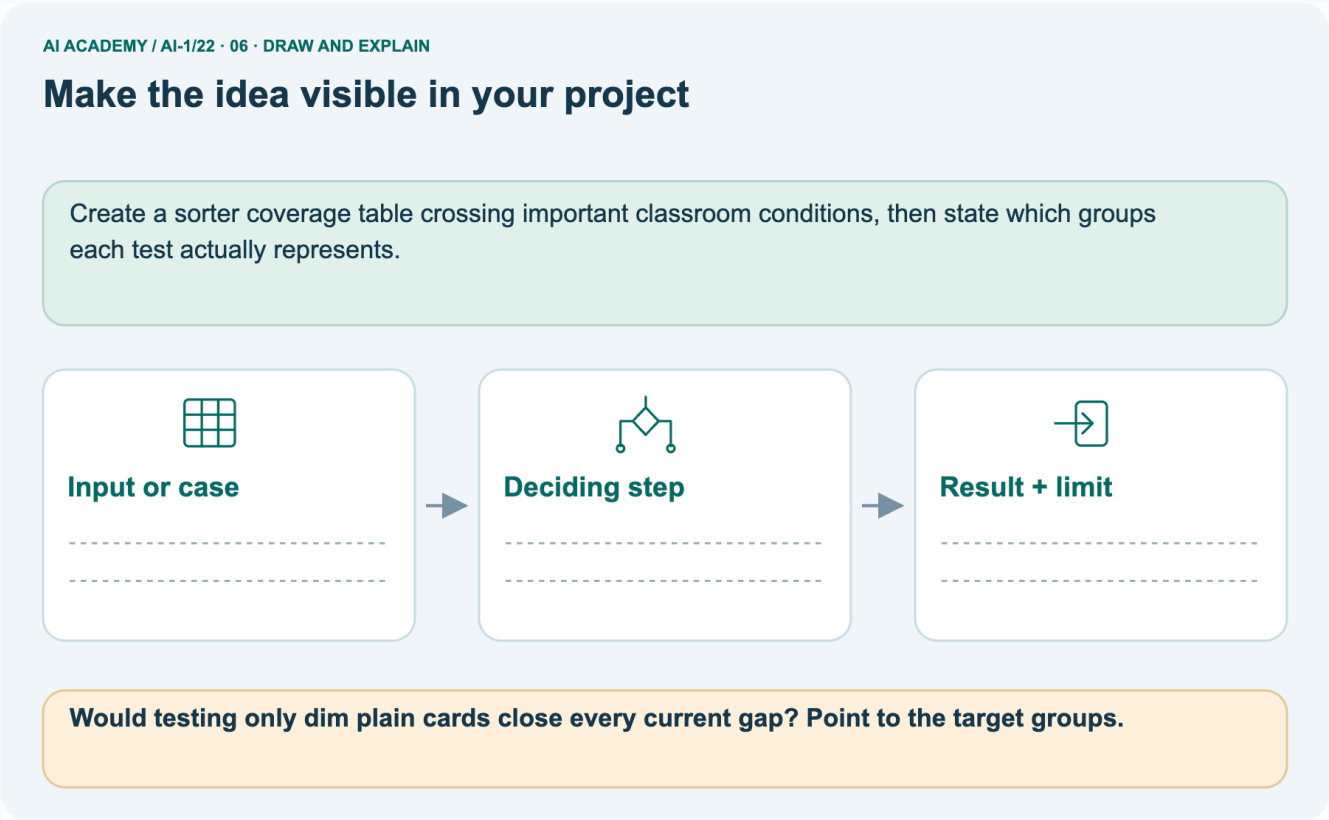
Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Limit: **Coverage** does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents. Draw your **input**, deciding step, result and limit.

Supported question: Would testing only dim plain cards close every current gap? Point to the target groups.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Missing and Unbalanced Examples: the case

02. Follow the deciding steps

03. Target groups and development coverage

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.